SIMATS ENGINEERING
ASSESSMENT TOOL 2

Name : E.Rohan
Reg No : 192524482

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 35%

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QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	5
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	5
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	5
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	5
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	5
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	5
7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	5

8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	**5**
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	**5**
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	**5**

Code of Conduct:

I E.Rohan-192524482 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

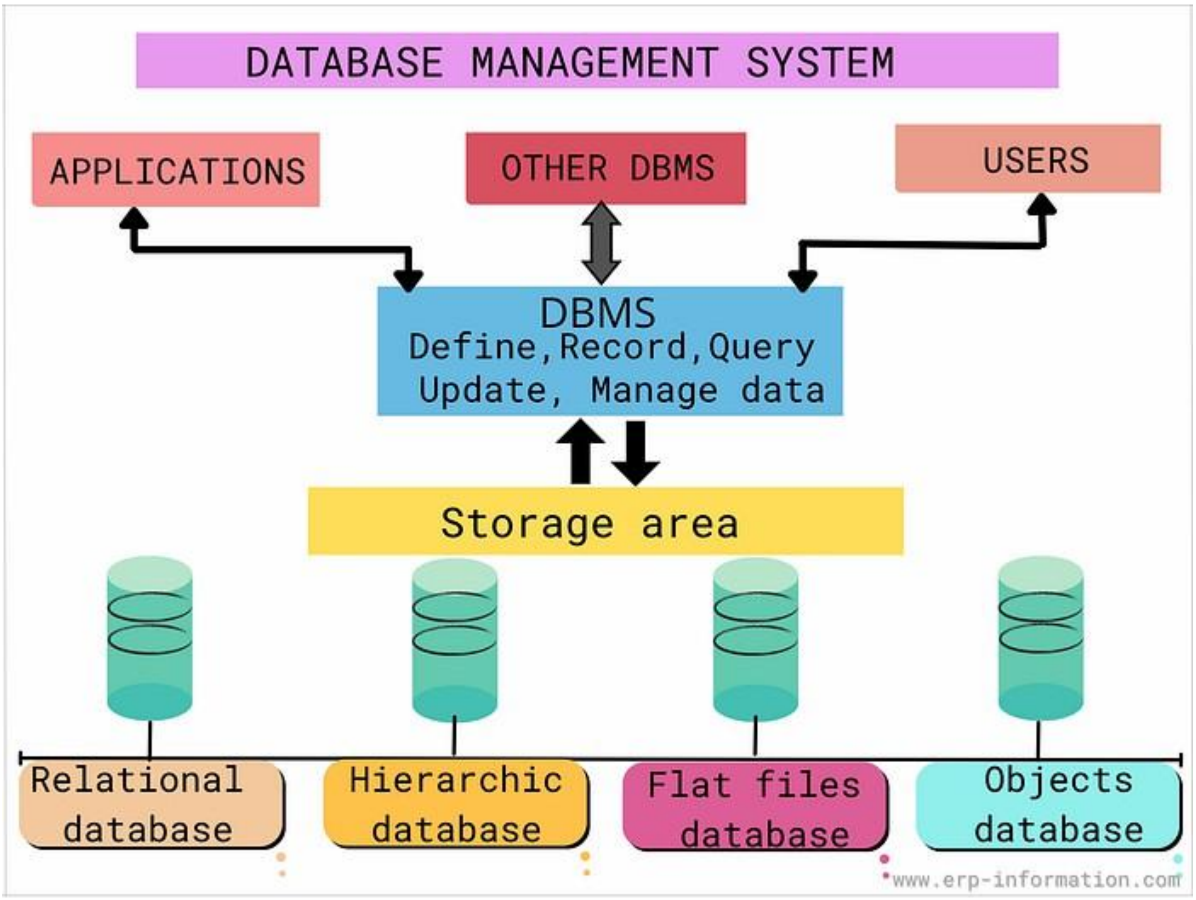
Signature of the student:

E.Rohan

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence

Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand



Q1.

ANS:

A DBMS is the software system that manages a database. It relies on a Data Model to decide how information is structured, and this structure is formally defined in the Schema. The Schema is described visually through an ER Diagram, which in turn is built from three basic building blocks: Entities, Attributes, and Relationships. At any given moment, the actual data held in the database is called an Instance — a snapshot of the schema populated with real values.

Key Linkages

- DBMS uses a Data Model → defines how data is organized.
- DBMS is defined by a Schema → the structural blueprint (tables, fields, constraints).
- DBMS has an Instance → the actual data present at a point in time.
- Data Model implements the Schema; Schema is populated as an Instance.
- Schema is represented visually by an ER Diagram.
- ER Diagram is composed of Entities, Attributes, and Relationships.

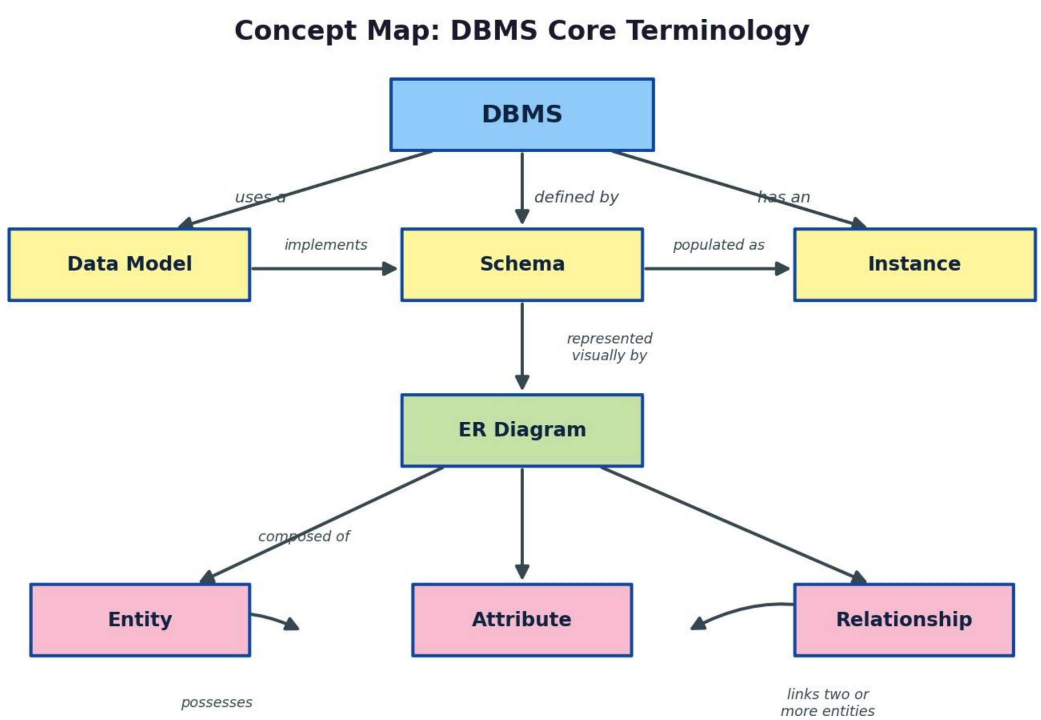


Fig 1.1 — Concept map of core DBMS terminology

Q2.

ANS:

The ANSI/SPARC architecture organizes a database into three levels so that changes at one level do not disturb the others. The External Level presents customized views to different users. The Conceptual Level gives one unified logical view of the entire database (entities, relationships, constraints), independent of physical storage. The Internal Level deals with how data is actually stored on disk — file organization, indexing, and access paths.

Data Independence

- Logical Data Independence: the conceptual schema can change (e.g. adding a new attribute) without altering external views/applications.
- Physical Data Independence: the internal/storage schema can change (e.g. new indexing, different file structure) without affecting the conceptual schema.

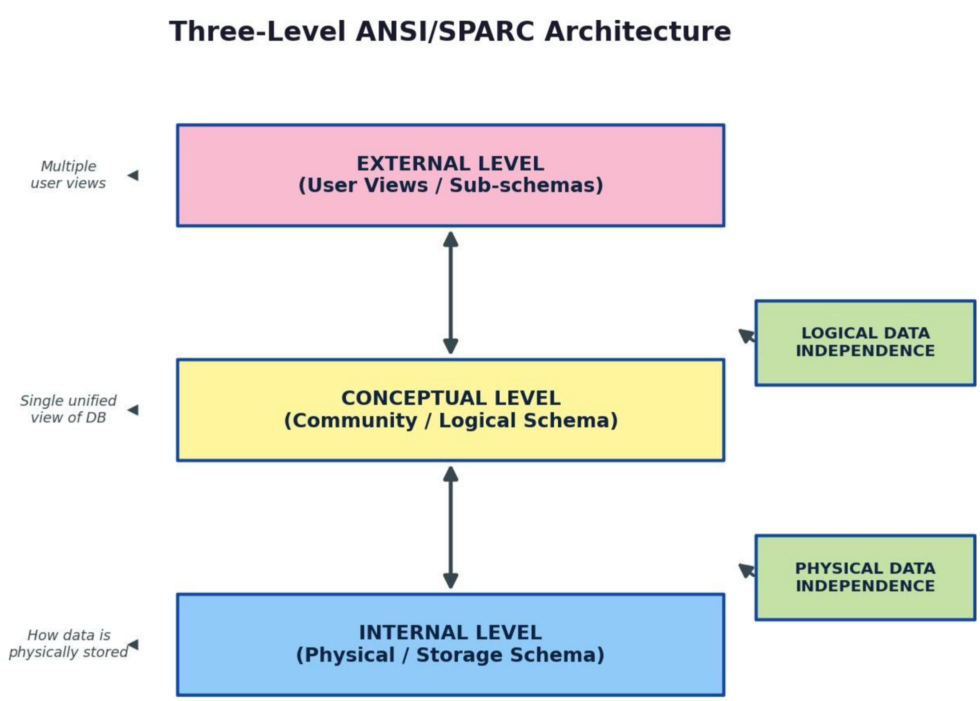


Fig 2.1 — Three-level architecture showing where each type of data independence applies

Q3.

ANS:

Before DBMSs, organizations stored data in separate files managed directly by application programs. This file-based approach caused a set of recurring problems, each of which a DBMS is specifically designed to solve.

Problems Solved by DBMS

- Data Redundancy & Inconsistency → solved by centralizing data so it is stored once and shared.
- Data Isolation (data scattered across files) → solved by an integrated data model accessible to all programs.
- Poor Data Integrity → solved by enforcing integrity constraints (e.g. primary/foreign keys, checks).
- Concurrent Access Anomalies → solved by concurrency control mechanisms (locking, timestamps).
- Security Weaknesses → solved by authentication and fine-grained access control.
- Lack of Atomicity/Recovery → solved by ACID transactions with backup and recovery mechanisms.

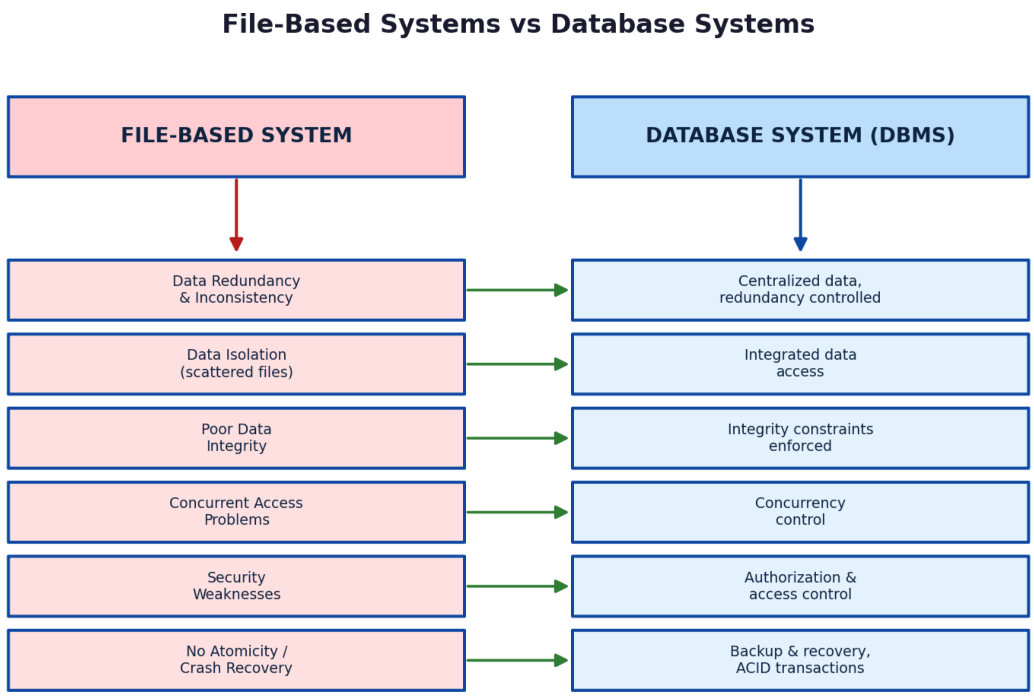


Fig 3.1 — Mapping of file-based system problems to their DBMS-based solutions

Q4.

ANS:

An ER diagram is built from four families of components. Entity Types can be Strong (independent, has its own key) or Weak (existence depends on an owner entity). Attribute Types include Simple, Composite, Single-valued, Multi-valued, Derived, and Key attributes. Relationship Types are classified by the number of participating entities: Unary/Recursive (one entity type relates to itself), Binary (two entity types), and Ternary (three entity types). Finally, Participation Constraints describe whether every entity instance must take part in a relationship (Total participation) or only some do (Partial participation).

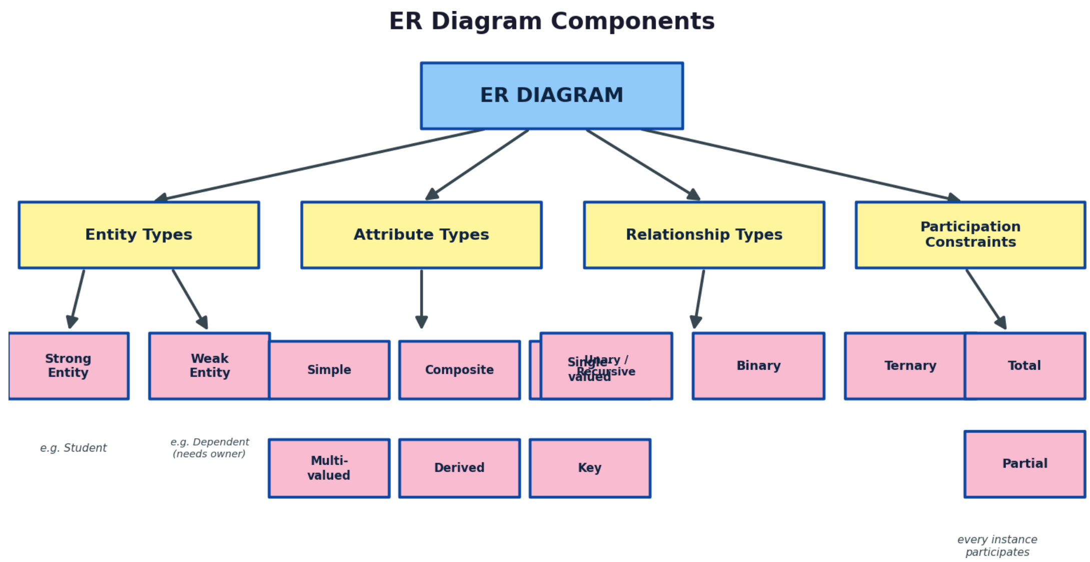


Fig 4.1 — Concept map of ER diagram component families

Q5.

ANS:

The three classical data models differ mainly in how they represent structure and relationships. The Hierarchical model arranges data as a tree with strict one-to-many parent-child links. The Network model generalizes this into a graph, allowing many-to-many relationships through owner-member sets. The Relational model, the most widely used today, represents everything as tables (relations) connected through keys, offering the greatest flexibility for querying via SQL.

Hierarchical vs Network vs Relational Data Models

	HIERARCHICAL	NETWORK	RELATIONAL
Structure	Tree (parent-child)	Graph (owner-member sets)	Tables of rows & columns
Relationships	One-to-many only	Many-to-many supported	1:1, 1:M, M:N via keys
Access Path	Fixed, via parent links	Set/pointer navigation	Flexible SQL queries
Example Use	File systems, XML data	Airline/CAD systems	Banking, ERP, most modern DBs

Fig 5.1 — Side-by-side comparison of the three data models

Q6.

ANS:

The Enhanced ER (EER) model adds three powerful modelling concepts on top of the basic ER model to capture more real-world semantics.

- Generalization: a bottom-up process that combines several similar entity types into one general superclass (e.g. Car and Truck generalized into Vehicle).
- Specialization: the reverse, top-down process — an entity type is divided into subclasses based on distinguishing characteristics (e.g. Vehicle specialized into Car and Truck).
- Aggregation: treats a relationship itself as a higher-level entity so that it can participate in further relationships (e.g. the Employee–WORKS-ON–Project relationship is aggregated so it can then be linked to a Manager).

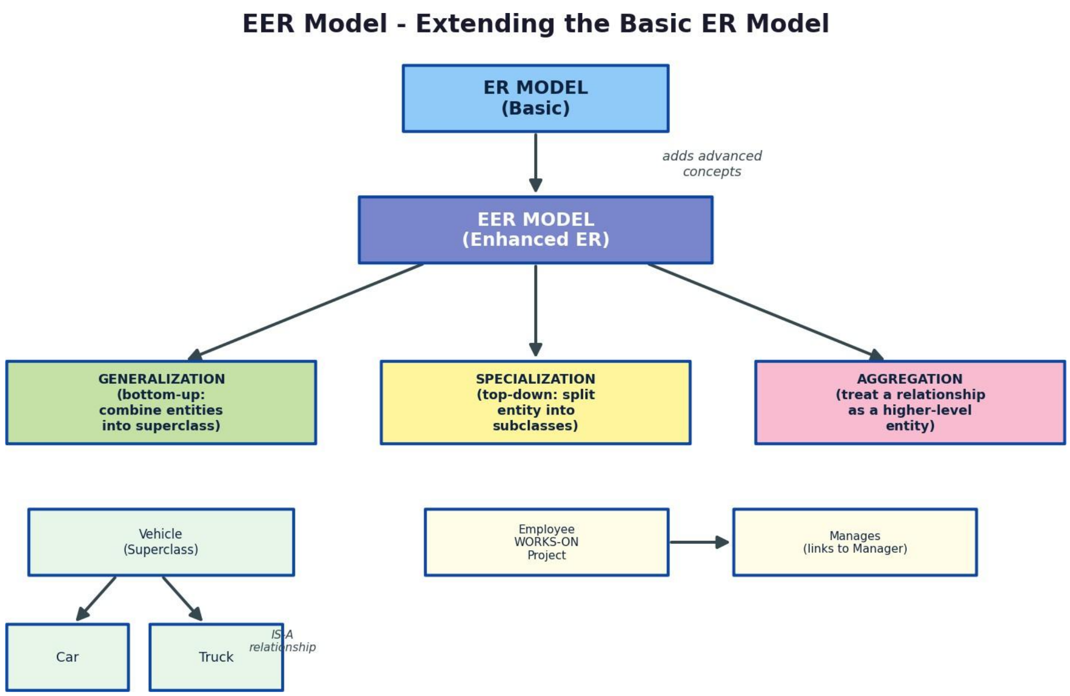


Fig 6.1 — EER concepts extending the basic ER model

Q7.

ANS:

When a user or application submits a query, it passes through a defined pipeline inside the DBMS before results are returned. The Query Processor first parses and validates the query syntax and semantics. The Query Optimizer then evaluates possible execution strategies and selects the most efficient plan. The Execution Engine carries out that plan, requesting data through the Storage/Buffer Manager, which manages data movement between disk and memory. Finally, the Physical Database on disk is accessed, and the result set travels back up through the same layers to the user.

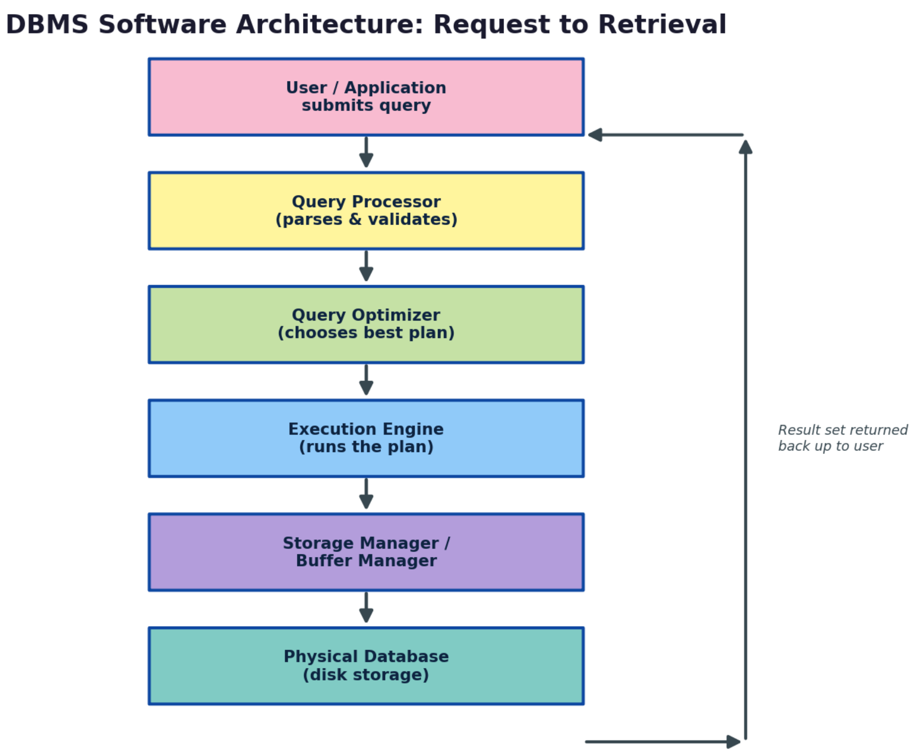


Fig 7.1 — DBMS software architecture: request-to-retrieval data flow

Q8.

ANS:

Each data model is best suited to particular kinds of applications, largely based on how naturally its structure fits the relationships in that domain.

- Hierarchical Model → File systems, XML documents, organization charts (naturally tree-shaped data).
- Network Model → Airline reservation systems, CAD/CAM engineering data (many-to-many links between records).
- Relational Model → Banking, ERP, e-commerce, and most modern applications (flexible querying, strong consistency).
- Object-Oriented Model → Multimedia databases, CAD, scientific applications (complex objects with behaviour).

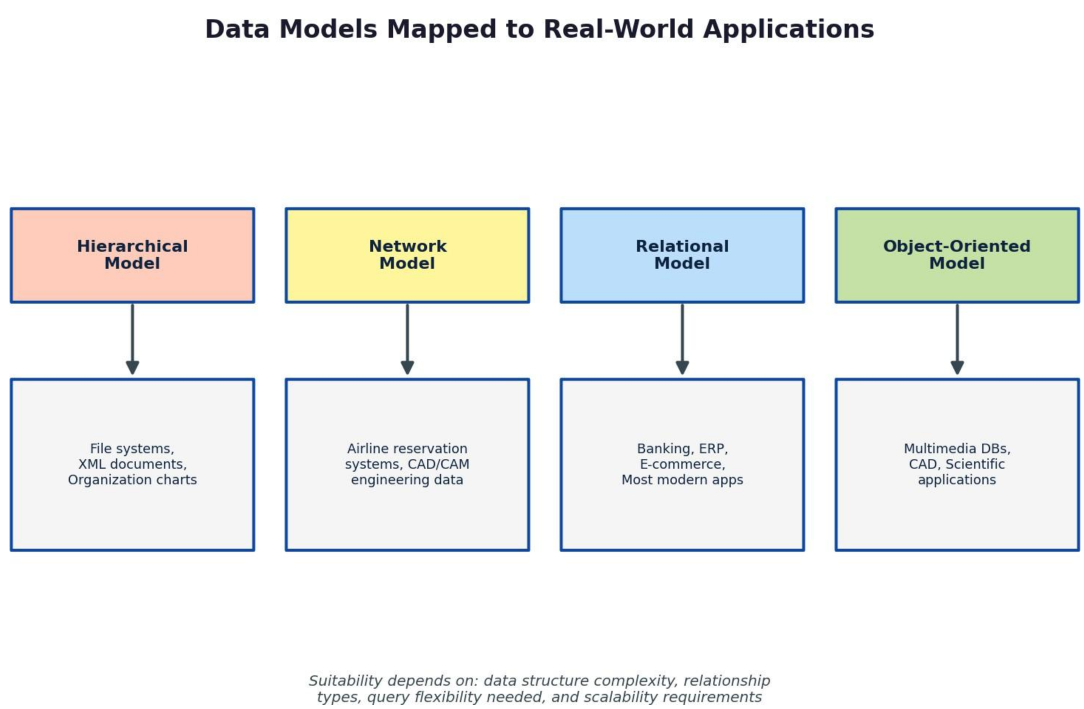


Fig 8.1 — Data models mapped to representative real-world applications

Q9.

ANS:

A Schema is the overall design or blueprint of a database — it is defined once and changes only occasionally. An Instance is the actual data stored at a particular moment in time — it changes frequently as records are inserted, updated, or deleted. These two concepts map directly onto the three schema levels: the External, Conceptual, and Internal schemas together define structure, while the data instance flows through them. Logical data independence protects external views when the conceptual schema changes; physical data independence protects the conceptual schema when the internal (storage) schema changes.

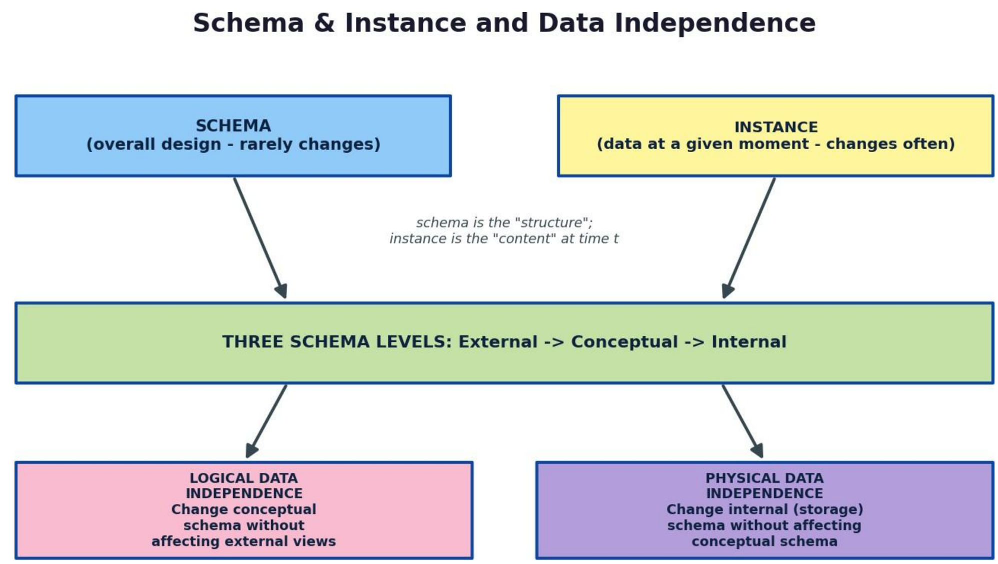


Fig 9.1 — Schema vs instance, connected to the three schema levels and data independence

Q10.

ANS:

A DBMS environment involves distinct roles that interact with the database differently.

- DBA (Database Administrator): responsible for schema design and modification, security and access control, and backup and recovery.
- Application Developer: writes programs that use the database, designs queries and transactions, and builds the user-facing applications.
- End User: naive users interact through forms/menus; sophisticated users may write SQL queries directly; most interact only via applications built by developers.

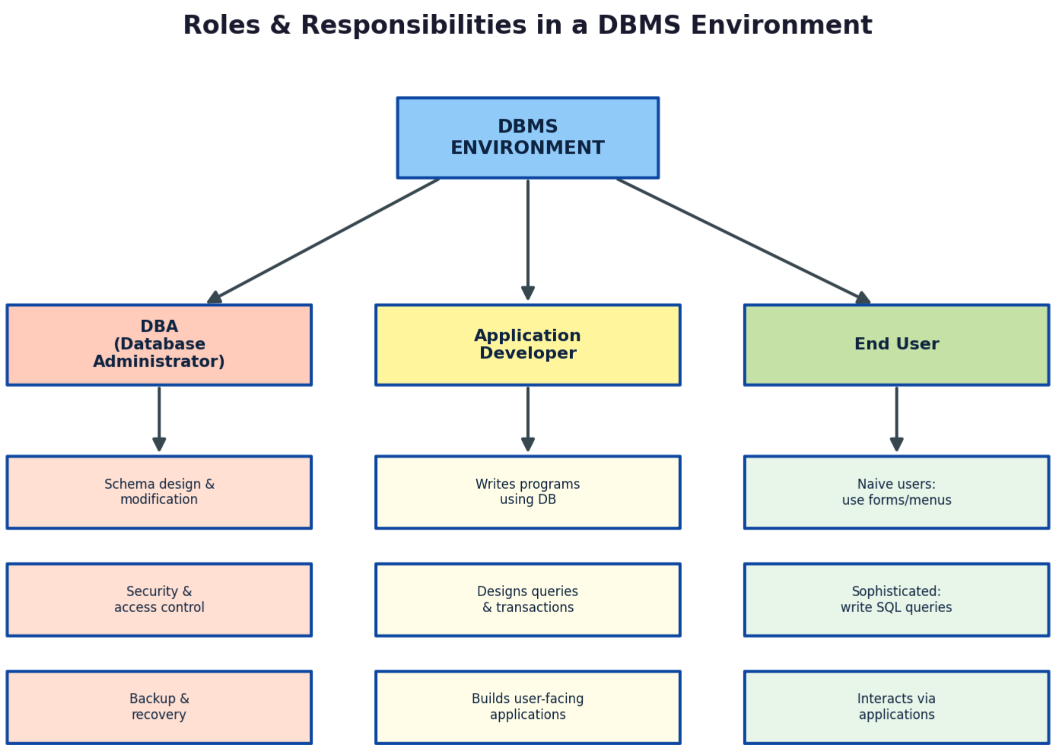


Fig 10.1 — Roles and responsibilities within a DBMS environment